Fiscal Year:	FY 2005	**Task Last Updated:**	FY 07/04/2005
PI Name:	Thorne, Robert E. Ph.D.		
Project Title:	Macromolecular Crystal Growth: Techniques Development, Fundamental Studies, and Applications to Complexes Affecting Cell Growth and Division		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Structural biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Ithaca	**State:**	NY
Zip Code:	14853-2501	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	08/30/2001	**End Date:**	11/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ellenberger, T. (Harvard Medical School) Duffy, T. (Geosciences Princeton University)		
Grant/Contract No.:	NAG8-1831		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Growth of high quality macromolecular crystals is a major factor limiting the rate at which new macromolecular structures are determined. We are pursuing an interdisciplinary experimental program to advance fundamental understanding of macromolecular crystals and to solve specific structures important in cell growth and division. Our program involves two major components: (1) Development and application of novel or improved experimental probes of macromolecular solutions and crystals, including X-ray imaging techniques, SAXS, and Brillouin scattering; and (2) Application of these and other techniques to help obtain high-quality crystals of DNA helicase-primase complexes and CDC42 and its regulators. The objectives of this work are the development of novel or improved experimental probes of macromolecular solutions and crystals, and specific attacks on crystal-related problems of molecular complexes involved in cell growth. New/improved techniques for characterizing macromolecular solutions and crystals should yield		

Task Description:	new insights into intermolecular interactions, crystal disorder, and molecular structure. Applying these and our current techniques (so far restricted to simple model proteins) to macromolecular complexes of significant biological interest will help to identify and refine a subset of these techniques most useful to the structural biology community. The graduate students involved in this program will receive an interdisciplinary training well suited to addressing future challenges in this expanding area. This program will assist NASA's microgravity effort by developing new methodologies for evaluating the success of flight experiments, by testing specific hypotheses about defect formation relevant to the microgravity environment, and by helping to disseminate the results from its microgravity crystal growth program.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work should significantly impact the general practice of macromolecular crystallography, and thus the fundamental understanding of biological processes and the development of new medicines.
Task Progress:	Master of Engineering student Guanhan Chew and Applied and Engineering Physics undergraduate student Matthew Sochor have been evolving existing designs and developing new kinds of tools for protein crystal manipulation and mounting. These include improved tools for crystal retrieval from crystallization media, for mounting multiple crystals, for mounting plates and rods, and for room-temperature crystallography on large crystals. We are developing a series of new platforms for conventional and high-throughput macromolecular crystal growth, one of which will be the subject of an upcoming STTR proposal. We have argued that motion of protein drops on crystallization media during routine handling is a major factor affecting the reproducibility of crystallization conditions. We have shown that drop stability can be enhanced by chemical patterning to more effectively pin the drop's contact line. As an example, a hydrophilic area is patterned on an initially flat hydrophobic glass slide. The drop remains confined to the hydrophilic area, and the maximum drop size that remains stable when the slide is rotated to the vertical position increases. This simple method is readily scalable and has the potential to significantly improve outcomes in hanging and sitting drop crystallization. Other innovations have so far only be described in patent disclosures. We have also completed a major review of our work on X-ray diffraction studies of protein crystal growth and disorder.
Bibliography Type:	Description: (Last Updated: 03/14/2017)
Articles in Peer-reviewed Journals	Kalinin, Y., Kmetko, J., Bartnik, A. Stewart, A. Gillilan, R., Lobkovsky, E., and Thorne, R. E. "A new sample mounting technique for room-temperature macromolecular crystallography." J. Appl. Cryst. 38 (2005) 333-339 , Feb-2005
Books/Book Chapters	Malkin, A. J., Thorne, R. E. "Growth and disorder of protein crystals: Insights from atomic force microscopy and X-ray diffraction studies" in "Methods, Volume 34" Ed. McPherson, A. J. Elsevier, 273-299 , Dec-2004